

SOC Audit Architect

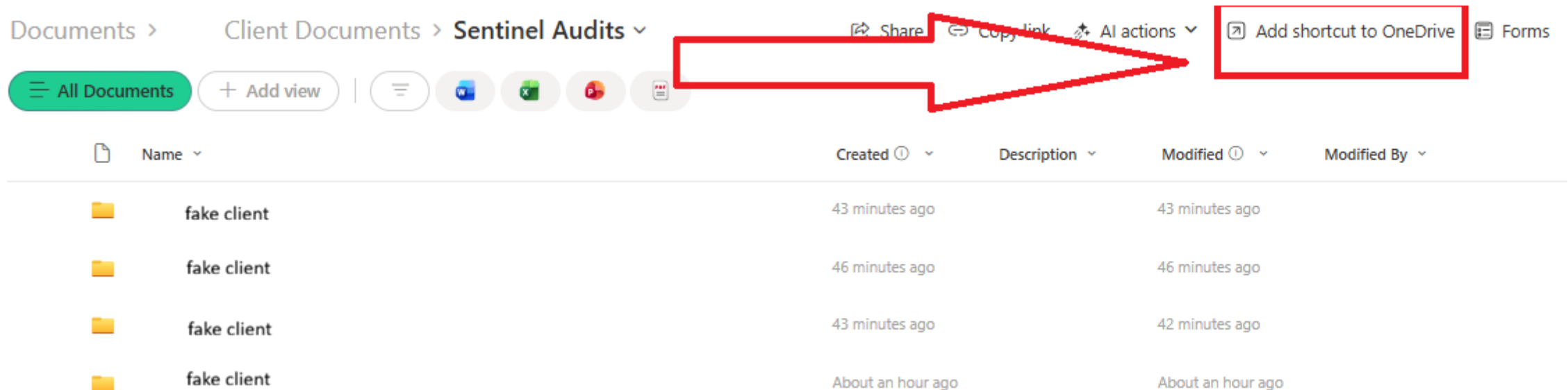
Automation for Sentinel Analytics Auditing.

Find the repository here:

<https://github.com/JamesHowardEGM/SOC-Audit-Architect/tree/main>

Step 1: Initialize the Environment

- The user establishes a shortcut to the OneDrive that the SOC shares.
- *Note: The script will ask if you have done this.*



The screenshot shows the Microsoft OneDrive interface. At the top, the breadcrumb navigation reads "Documents > Client Documents > Sentinel Audits". Below this is a toolbar with buttons for "All Documents", "Add view", and application icons for Word, Excel, PowerPoint, and OneNote. To the right of these icons are buttons for "Share", "Copy link", "AI actions", "Add shortcut to OneDrive", and "Forms". A red arrow points from the "Share" button to the "Add shortcut to OneDrive" button, which is also enclosed in a red rectangular box. Below the toolbar is a table with columns: "Name", "Created", "Description", "Modified", and "Modified By". The table contains four rows, each representing a folder named "fake client".

Name	Created	Description	Modified	Modified By
fake client	43 minutes ago		43 minutes ago	
fake client	46 minutes ago		46 minutes ago	
fake client	43 minutes ago		42 minutes ago	
fake client	About an hour ago		About an hour ago	

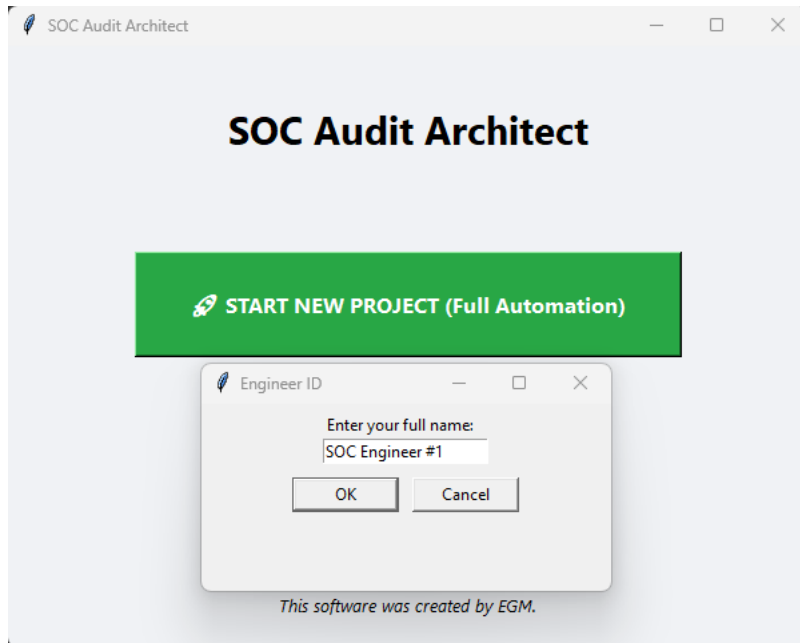
Step 2: Download Master JSON from Sentinel

- 1. Select the rules you want to audit.
- 2. Export them;
a JSON file will automatically download into your "Downloads" folder

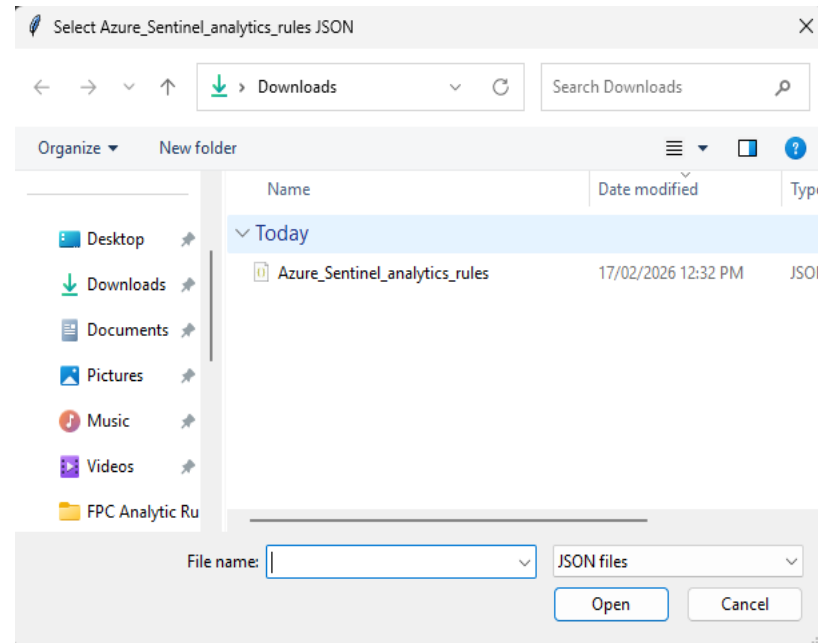
The screenshot shows the Microsoft Sentinel 'Active rules' page. At the top, there's a toolbar with buttons like 'Create', 'Refresh', 'Analytics workbooks', 'Rule runs (Preview)', 'Enable', 'Disable', 'Delete', 'Import', and 'Export'. The 'Export' button is highlighted with a red box and a red '2.' next to it. Below the toolbar, there's a summary section showing '178 Active rules' and a 'Rules by severity' bar chart with categories: High (63), Medium (104), Low (13), and Informational (1). The main section is titled 'Active rules' and includes a search bar and an 'Add filter' button. A table of rules is displayed below, with columns for 'Rule type' and 'Status'. The first 12 rules are highlighted with a red box, indicating they are selected for export. These rules are all 'Medium' severity and 'Scheduled' rule type. The first rule is 'Phishing link click observed in Network Traffic'.

Severity	Rule name	Rule type	Status
Medium	Phishing link click observed in Network Traffic	Scheduled	Enabled
Medium	TI map IP entity to DNS Events (ASIM DNS schema)	Scheduled	Enabled
Medium	TI Map URL Entity to UrlClickEvents	Scheduled	Enabled
Medium	TI map File Hash to Security Event	Scheduled	Enabled
Medium	TI map Domain entity to Dns Events (ASIM DNS Schema)	Scheduled	Enabled
Medium	TI map IP entity to Network Session Events (ASIM Network Session schema)	Scheduled	Enabled
Medium	TI map IP entity to Cloud App Events	Scheduled	Enabled
Medium	TI map URL entity to Cloud App Events	Scheduled	Enabled
Medium	TI map Domain entity to SecurityAlert	Scheduled	Enabled
Medium	TI Map Domain Entity to DeviceNetworkEvents	Scheduled	Enabled
Medium	TI map Domain entity to Cloud App Events	Scheduled	Enabled
Medium	TI map Email entity to Cloud App Events	Scheduled	Enabled
Medium	TI map Domain entity to DnsEvents	Scheduled	Enabled

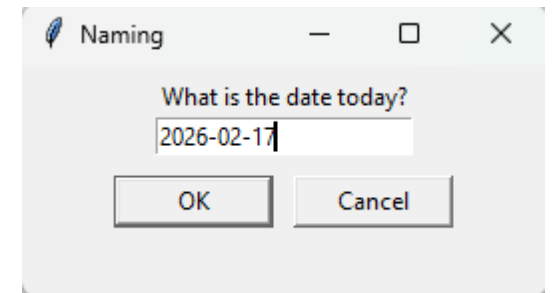
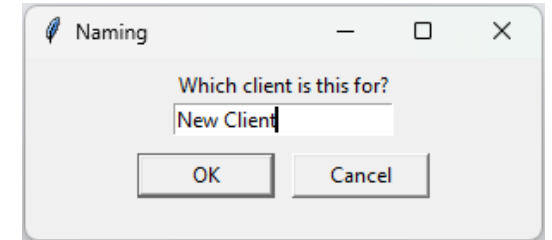
Step 3: Run SOC-Audit-Architect



- After clicking start new project;
- Enter your full name.



- Select the JSON file you downloaded from Sentinel



- Name the Client
- Confirm the date (it's automatic)

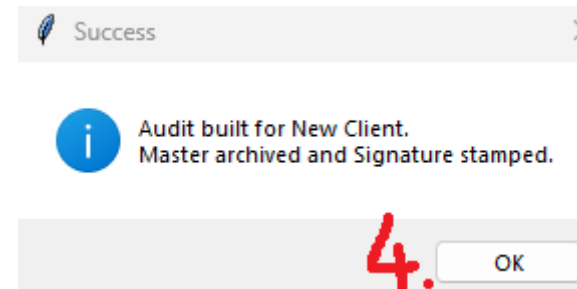
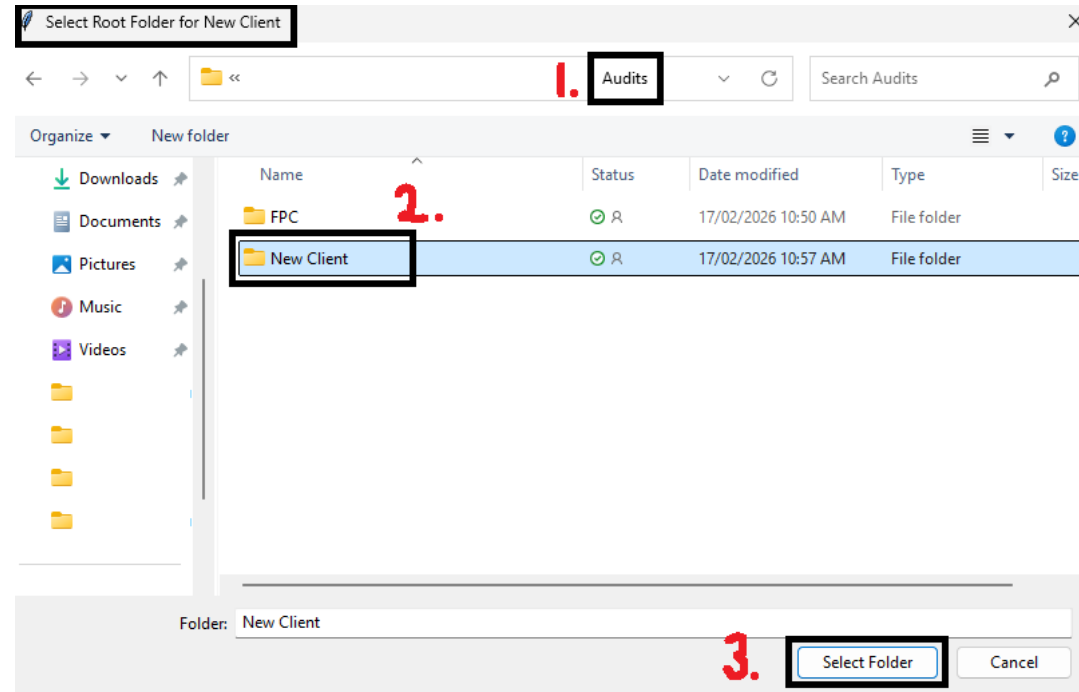
Step 4: Selecting the Root directory for the client (see step 1)

1. Go to your Root Audits folder

2. Select the Client you're auditing

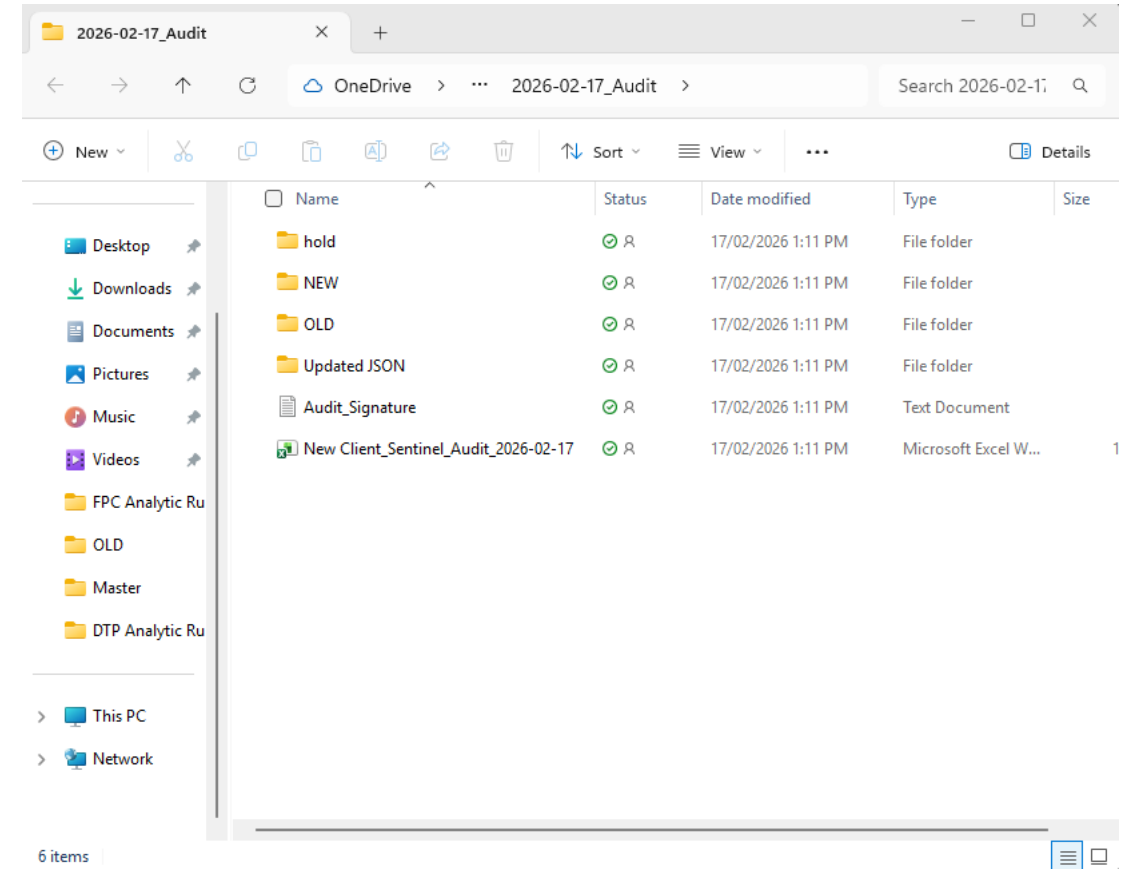
3. Confirm the client by clicking "Select Folder"

4. Confirm that the Automation is successful, and click "OK"



Step 5: Understanding Automated Directories

- Upon completing Step 4; The Audit Directory will automatically open, and you will see something like this:
-  OLD/
- JSON files containing the pre-update versions of the rules.
-  NEW/
- JSON files containing the updated versions of the rules.
-  hold/
- Rules that still require automation or testing before they can be finalized.
-  Updated Json/
- Folder used to send updated rule files through DevOps for deployment.
-  Analytics Rules Table (Excel Document)
- Research, analysis, and findings related to analytics rules.
-  Audit Signature
- A simple document recording who is making the audit file



Step 5.1: Understanding Automated Directories – The OLD folder.

- Remember Sentinel? When you Enter into the “OLD” folder, You’ll see that the automated program has divided your master JSON file into all the individual rules that you selected from Sentinel!

☒	Medium	UPDATE	TI map IP entity to DNS Events (ASIM DNS schema)
☒	Medium	UPDATE	TI Map URL Entity to UrlClickEvents
☒	Medium	UPDATE	TI map File Hash to Security Event
☒	Medium	UPDATE	TI map Domain entity to Dns Events (ASIM DNS Schema)
☒	Medium	UPDATE	TI map IP entity to Network Session Events (ASIM Network Session schema)
☒	Medium	UPDATE	TI map IP entity to Cloud App Events
☒	Medium	UPDATE	TI map URL entity to Cloud App Events
☒	Medium	UPDATE	TI map Domain entity to SecurityAlert
☒	Medium	UPDATE	TI Map Domain Entity to DeviceNetworkEvents
☒	Medium	UPDATE	TI map Domain entity to Cloud App Events
☒	Medium	UPDATE	TI map Email entity to Cloud App Events
☒	Medium	UPDATE	TI map Domain entity to DnsEvents

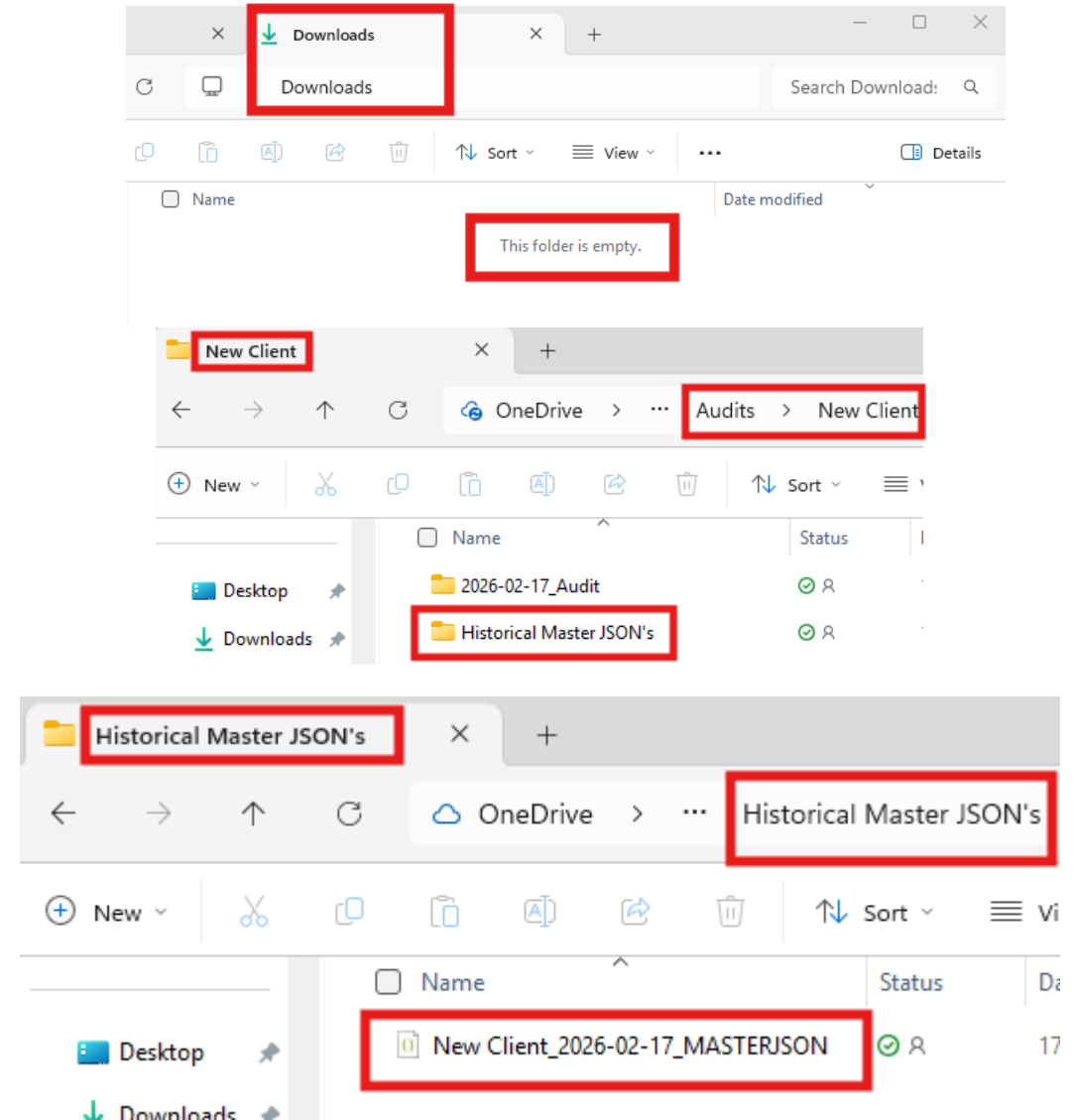


Name	Status	Date modified	Type	Size
TI map Domain entity to Cloud App E...	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI Map Domain Entity to DeviceNetwo...	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map Domain entity to Dns Events (...)	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map Domain entity to DnsEvents	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map Domain entity to SecurityAlert	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map Email entity to Cloud App Eve...	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map File Hash to Security Event	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map IP entity to Cloud App Events	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map IP entity to DNS Events (ASIM ...)	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map IP entity to Network Session Ev...	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI map URL entity to Cloud App Events	✓ R	17/02/2026 1:11 PM	JSON Source File	
TI Map URL Entity to UrlClickEvents	✓ R	17/02/2026 1:11 PM	JSON Source File	

Step 5.2: Understanding Automated Directories – The MASTER JSON file

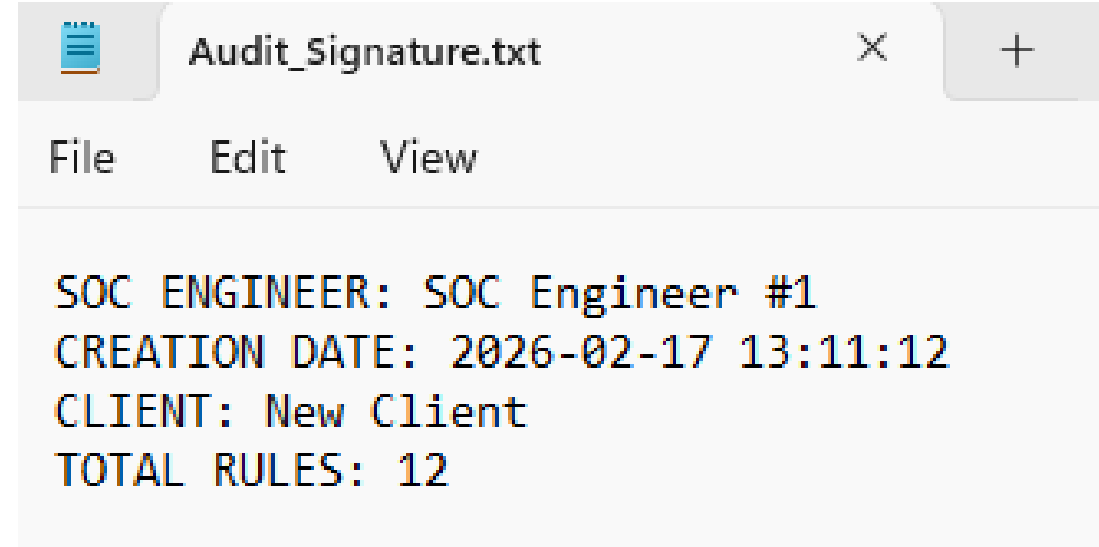
- Now that we have all the individual rules extracted from the master JSON.
- What happened to the original file?
- The automated program will move the master JSON from the download folder, into a directory that collects all the master files, titled, “Historical Master JSON’s”
- Finally, when you enter that folder, you’ll see that the automated program renamed the Master JSON file to a standard format of:

• `[client name] [date of audit] [MASTERJSON]`
- *This is all automated. You don’t have to worry about it. – Your download folder doesn’t get clogged up, and your auditing file system is automatically updated .*



Step 6: Understanding Automated Files – The “Audit_Signature.txt” file

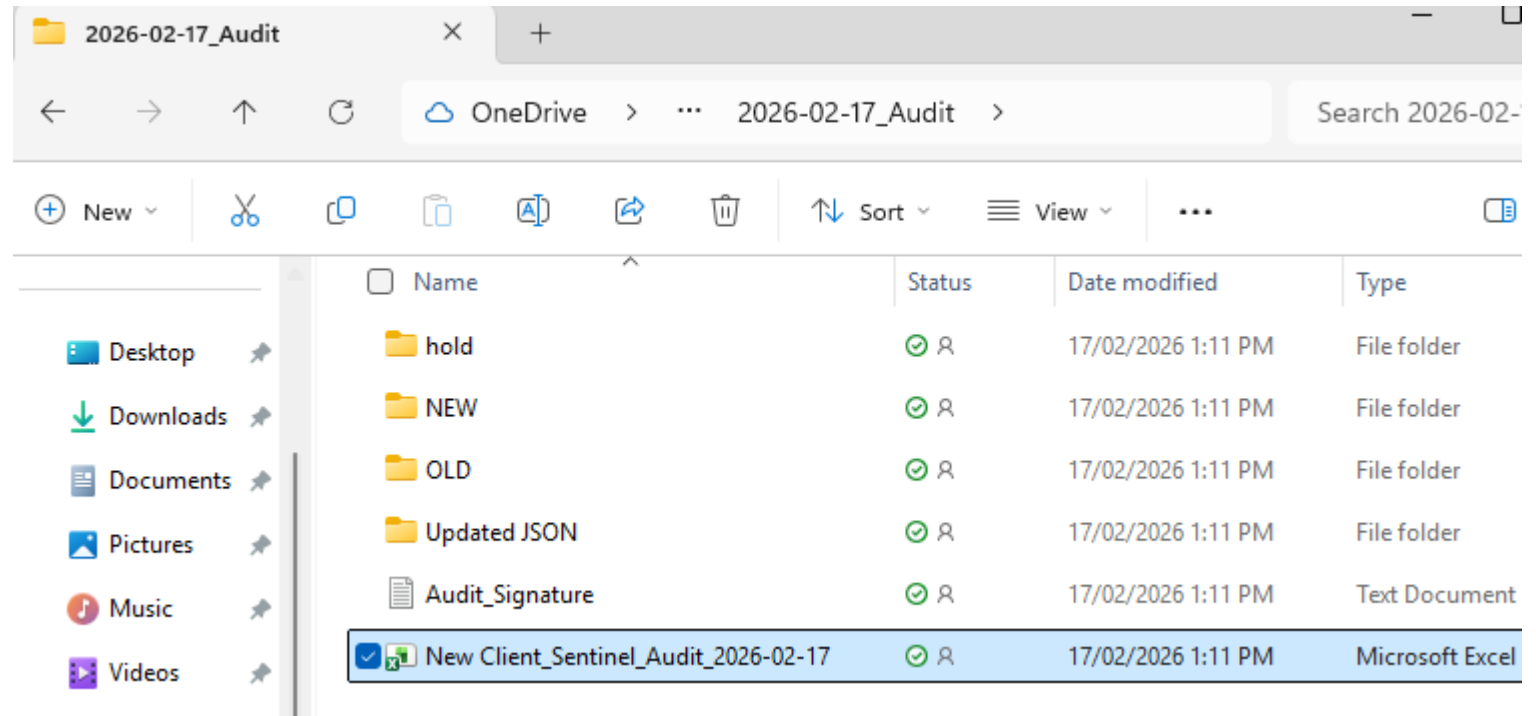
- Inside the Audit file, you’ll see a text document titled, “Audit Signature”
- This will tell you information about who did this audit, what date, the client and how many rules were audited

A screenshot of a text editor window. The title bar at the top shows a blue notepad icon, the filename 'Audit_Signature.txt', a close button (X), and a plus sign (+). Below the title bar is a menu bar with 'File', 'Edit', and 'View' options. The main text area contains the following text:

```
SOC ENGINEER: SOC Engineer #1  
CREATION DATE: 2026-02-17 13:11:12  
CLIENT: New Client  
TOTAL RULES: 12
```

Step 6.1: Understanding Automated Files – The Excel Spreadsheet File.

- This is where the real magic happens.
- This function saves so much time.
- Open the Excel document to view the power of automation!



Step 6.2: Understanding Automated Files – The Excel Spreadsheet File.

The automation tool will create a spread sheet, and import the Rule Name, Current Version, Verify if its Enable/Disabled and extract the Current Rule into the “Old KQL” column.

Just for this demonstration I've only shown a small part of what else is included in the generated spread sheet. Continue to the next page to see more!

Rule Name	Current Version	Newest Version	Last Modified	Enabled/Disabled	Changes/Modifications	Notes	Comments	Old KQL
TI map IP entity to DNS Events (ASIM DNS schema)	1.2.6			Enabled				<pre>let dt_lookBack = 1h; let ioc_lookBack = 14d; let IP_TI = ThreatIntelIndicators where TimeGenerated >= ago(ioc_lookBack) //extract key part of kv pair extend IndicatorType = replace(@"\[\] \\\""", "", toString(split(ObservableKey, ":", 0))) where IndicatorType == "ipv4-addr" extend TrafficLightProtocolLevel = toString(parse_json(AdditionalFields).TLPLLevel) extend NetworkSourceIP = ObservableValue extend IoC = NetworkSourceIP summarize LatestIndicatorTime = arg_max(TimeGenerated, *) by Id, ObservableValue where IsActive and (ValidUntil > now() or isempty(ValidUntil)); IP_TI project-reorder *, IsActive, Tags, TrafficLightProtocolLevel, NetworkSourceIP, Type, IoC join kind=innerunique // using innerunique to keep perf fast and result set low, we only need one match to indicate potential malicious activity that needs to be investigated (_Im_Dns(starttime=ago(dt_lookBack)) where isempty(DnsResponseName) summarize imDns_mintime=min(TimeGenerated), imDns_maxtime=max(TimeGenerated) by SrcIpAddr, DnsQuery, DnsResponseName, Dns_EventProduct, EventVendor</pre>

Step 6.3: Understanding Automated Files – The Excel Spreadsheet File.

- This is what the other columns look like in the generated spreadsheet. Notice how there is a column for ‘Audit Judgement’. There is a drop-down menu that allows you to select a category for future reference.
- If you open the excel document, you will see that there will be a row representing every Rule that you selected in Sentinel.

Old KQL	Updated KQL	Changes	Test Results	Notes	Before	After	Audit Judgment
<pre>let dt_lookBack = 1h; let ioc_lookBack = 14d; let IP_TI = ThreatIntelIndicators where TimeGenerated >= ago(ioc_lookBack) //extract key part of kv pair extend IndicatorType = replace(@"\[\\ \\\""", "", tostring(split(ObservableKey, ":", 0))) where IndicatorType == "ipv4-addr" extend TrafficLightProtocolLevel = tostring(parse_json(AdditionalFields).TLPLLevel) extend NetworkSourceIP = ObservableValue extend IoC = NetworkSourceIP summarize LatestIndicatorTime = arg_max(TimeGenerated, *) by Id, ObservableValue where IsActive and (ValidUntil > now() or isempty(ValidUntil)); IP_TI project-reorder *, IsActive, Tags, TrafficLightProtocolLevel, NetworkSourceIP, Type, IoC join kind=innerunique // using innerunique to keep perf fast and result set low, we only need one match to indicate potential malicious activity that needs to be investigated (_Im_Dns(starttime=ago(dt_lookBack)) where isnotempty(DnsResponseName) summarize imDns_mintime=min(TimeGenerated), imDns_maxtime=max(TimeGenerated) by SrcIpAddr, DnsQuery, DnsResponseName, Dns_EventProduct, EventVendor)</pre>							Default
							Default Update on Hold No. of alerts same before/after OSG Modified Rules Urgent Attention Needed